

Machine learning practice questions

1. Consider a single data point $x = 2, y = 10$ and a linear model

$$\hat{y} = wx$$

The loss is

$$L(w) = \frac{1}{2}(wx - y)^2$$

If current weight is $w = 1$ and learning rate is $\eta = 0.1$:

- Compute the gradient $\frac{\partial L}{\partial w}$
- Perform **one step** of gradient descent to find the updated weight w_{new}
- Explain whether a larger η would risk divergence.

2. You collect the following data of building height vs wind load,

Height (m)	Load (kN)
10	18
20	31
30	45
40	62

You try the following models:

- Model A: linear $WL = a + bh$
- Model B: cubic $WL = c_0 + c_1h + c_2h^2 + c_3h^3$

- Which model is more prone to overfitting? Why?
- Explain how you would detect overfitting.

3. Suppose two features are used to predict material strength:

Predictor 1, x_1	Predictor 2, x_2	Output, y
2000	5	40
2100	7	38
2300	9	35

- Compute mean and standard deviation of each predictor
- Normalize the predictor values to zero mean and unit variance ($x_{norm} = \frac{x - \mu_x}{\sigma_x}$)
- Explain why we need normalization

4. You are given the following data for predicting concrete compressive strength from water–cement ratio:

Water-cement ratio, x	Strength, y
0.5	42
0.6	38

You assumed linear model, $y = a + bx + \epsilon$

- Write design matrix A and observation matrix y

- b) Compute least squares estimate $\hat{\beta} = (A^T A)^{-1} A^T y$
- c) Find mean squared error loss using these optimal parameters

5. You want to classify a new soil sample based on density and moisture.

Density	Moisture	Soil type
1.0	6	Clay
1.5	4	Sand
2.0	8	Clay
2.5	5	Sand

- a) What is the type of task you are trying to solve (classification, regression or clustering) ?
- b) You now are given a new point, density = 1.7 and moisture = 6. Compute Euclidean distance to all points.
- c) What is the soil type if you use KNN with k=1?
- d) What is the soil type if you use KNN with k=3?

6. You have a dataset of 120 samples. You split it as: 70% training, 15% validation, 15% testing.

- a) How many samples go into each set?
- b) Explain the purpose of the validation set.
- c) What happens if the test set is used to tune hyperparameters?

7. You observe the following validation errors after training the linear regression model with different learning rates:

Learning rate	Validation Error
0.001	12
0.01	6
0.1	1200

- a) Do these validation errors make sense to you? Why or why not?
- b) What is the learning rate you would use for training?
- c) What do you classify learning rate as? Model parameter or hyperparameter? Why?

8. You attempt a polynomial regression of degree d to predict load from displacement. The measured errors are:

Degree, d	Train error	Validation error
1	12	14
2	6	7
5	1	20

- a) Which model suffers from high bias?
- b) Which model suffers from high variance?
- c) Which model should you pick and why?

- d) You get a second set of data from experiments. In which model will the parameters be very different from the earlier data?

9. Given data:

$$x = [1, 2, 3] \text{ and } y = [2, 2.5, 3.5]$$

- Compute closed form Least Squares estimate assuming a model of the form $y = bx$
- If initial guess is $b_0 = 0$, find the gradient descent update after 1 iteration with learning rate = 0.1
- Why do you think we would use gradient descent even for least-squares estimate for large scale data.

10. You want to predict compressive strength of concrete by using the water-cement ratio.

w/c ratio	Strength (MPa)
0.4	55
0.52	42
0.6	36
0.65	33

You are now asked to predict the strength of concrete with water-cement ratio of 0.58. You decide to use kNN regression.

- What is your estimate for $k=1$?
- What is your estimate for $k=2$?
- Which kNN tends to overfit more? $k=1$ or $k=5$? Why?

11. You are fitting polynomial models to the following dataset:

Example Number	x	y
1	0	1
2	1	2
3	2	2
4	3	3

You decide to use quadratic model: $y = c_0 + c_1x + c_2x^2$. You need some data for validation, so you decide to keep one data point out. You try out two models. In the first model, you use the 1st, 2nd and 3rd examples for training and 4th for validation. In the second model, you use the 2nd, 3rd and 4th for training and 1st for validation. Find the least squares estimate for parameters of these two models. Explain why they are so different.